

11 CAUSES AND ACTIONS FOR ERRORS AND FAILURES

11.1 Summary

This chapter describes the causes of abnormality and malfunction and corrective action.

When the weighing machine fails and an error message is displayed on the remote control unit, refer to "11.2 Error Display and Action". When the weighing machine fails but no error message is displayed, refer to "11.3 Failure and Malfunction of the Weighing Machine".

For further technical actions, refer to "11.4 Troubleshooting".

If unable to discover the cause or appropriate action for the abnormality or malfunction, or when the weighing machine does not return to normal condition, contact the distributor or Ishida customer support.

DANGER

- **All inspection and servicing of the main body, control panel or operation panel can be performed only by Ishida service department. Contact the distributor or Ishida Customer Support.**

WARNING

- **For inspection and repair, unless instructed, the operator must turn OFF and lock the main power switch, and keep the key in his possession during the work (Refer to "1.6 Power Supply Shut Off and Indication").**
- **When performing inspection or repair, clearly indicate to other operators that inspection and repair are in progress.**
- **When performing inspection and repair for the main body of the weighing machine, turn OFF the main power switch, and wait at least five minutes before starting the work.**
- **When performing inspection and repair for the upper part of the weighing machine, use a proper scaffold or stepladder in order to avoid a fall.**

<Contents>

- Error display and action
- Malfunction of weighing machine, cause and action

<Purpose>

- To understand and master the handling method and action for the trouble experienced during operation, in order to enhance the operating efficiency of the weighing machine.

<Intended reader>

- System administrators
- Maintenance engineers

11.2 Error Display and Action

If an error occurs during operation, draining, or zero adjustment, an error message is displayed on the screen of the remote control unit.

There are eight error message screen patterns depending on the display method, alarm sound or weighing machine status (Refer to "Table 11-1").

Actions for errors differ depending on the error screen pattern. Check the screen pattern of the error and refer to details of each pattern described in "11.2.1 Pattern 1" through "11.2.8 Pattern 8", to cancel the error display and perform the correcting action.

Table 11-1

Error Screen Pattern	Error Name	Weighing Machine Condition
Pattern 1	• [ZERO ERROR] (Auto zero adjustment) • [OVERWEIGHT ERROR] • [OVERSCALE ERROR] • WH, PH, RS, TH, and DTH ERROR • [PIECE WEIGHT ERROR] (Count set weighing specification)	• Error alarm sounds • Operation stops Refer to "11.2.1 Pattern 1"
Pattern 2	• [ZERO ERROR] (Manual zero adjustment) • [SPAN ERROR] • [NETWORK ERROR] • [RECHECK ERROR] (Count set weighing specification) • [FDRV ERROR]	• Error alarm sounds • Operation stops Refer to "11.2.2 Pattern 2"
Pattern 3	• [WARNING:NO TRANSMISSION] • [WARNING:NO RESPONSE][LOW PRODUCT] • [WARNING:OVERSCALE] • [WARNING:DETACHED HEAD] • [WARNING:OVERWEIGHT ERROR] • [RCU fan stopped. Contact to factory service.] • [WARNING:AUTO ZERO ERROR] • [WARNING:COMMUNICATION ERROR] • [WARNING:RECHECK ERROR] • WARNING:PH and WH ERROR	• Error alarm sounds Refer to "11.2.3 Pattern 3"
Pattern 4	• [The memory of <Product Set> may be corrupt.] • [The memory of <Weigh Spec. Set> may be corrupt.] • [The memory of <Combi.Calc.Spec.Set> may be corrupt.] • [The memory of <Sectioning Method Setting> may be corrupt.] • [The memory of <Infeeder Control SET> may be corrupt.] • [The memory of <Packer Interlock Setting> may be corrupt.] • [COPY/Initialize error]	• Operation stops Refer to "11.2.4 Pattern 4" This error is displayed only when weighing machine operation is stopped.
Pattern 5	• [AFV ZERO ERROR] • [AFV PARAMETER ERROR] • [AFV COMPENSATION ERROR] • [AFV OUTPUT ERROR] • [AFV OVERFLOW ERROR]	• Error alarm sounds • Operation stops • Power OFF Refer to "11.2.5 Pattern 5"

Table 11-1

Error Screen Pattern	Error Name	Weighing Machine Condition
Pattern 6	• [Total Overflow!]	• Operation stops Refer to "11.2.6 Pattern 6"
Pattern 7	• [ADC firmware error]	• Operation stops • Power OFF Refer to "11.2.7 Pattern 7"
Pattern 8	• [FILE TRANSFER ERROR]	• Error alarm sounds Refer to "11.2.8 Pattern 8"
Pattern 9	• [Can not access Master Data!] • [No data in Master Memory]	• Alarm sounds Refer to "11.2.9 Pattern 9"
Pattern10	• [RCU fan stopped. Contact to factory service.]	• Alarm sounds Refer to "11.2.10 Pattern 10"

11.2.1 Pattern 1

If an error occurs during operation, draining, or zero adjustment, warning alarm sounds and the weighing machine stops automatically. In addition, error name, error location, error description and result, cause of the error and its action are displayed on the screen of the remote control unit.

When a pattern 1 error occurs, refer to "Table 11-2 " to eliminate the cause of the error and clear the error.

Table 11-2

Error Name/ Error Description	Cause	Action
[ZERO ERROR]	1. Weighed product is remaining in weigh hopper. 2. Weighed product is spilling from the pool hopper. TIP This error occurs when the zero adjustment exceeds the zero adjustment permissible limit due to the above cause.	After checking the following points, perform the manual zero adjustment. 1. Remove the products remaining in the weigh hopper. 2. Inspect for looseness, breakage or loss of the screws of the weigh hopper. If unable to repair, contact the distributor or Ishida customer support to request repair service. 3. If repair cannot be achieved by the above actions 1 and 2, there is a possibility of malfunctioning of the ADC board or power. Contact the distributor or Ishida customer support. Refer to "Table 11-13 "

Table 11-2

Error Name/ Error Description	Cause	Action
[OVERWEIGHT ERROR] /Proper combination was not obtained from calculation.	1.Excessive product fed into weigh hopper. 2.Excessive product fed into dispersion table or dispersion feeder. 3.The set value of the upper weight limit is small. TIP This error occurs when a proper result of combination weighing cannot be obtained and all combination weights are underweight or overweight, or when overweight occurs repeatedly due to vibration, etc. for the set number of times during rechecking. If overweight is set to be automatically discharged, production will not be stopped due to this error, and a warning message will be displayed on the lower side of the screen of the remote control unit.	1.If 1-3 heads are selected for combination, the amount of supplied product is excessive. In this case, decrease the feeder amplitude or time. 2.In order to prevent an excessive supply of products, decrease the amount of products supplied from the infeed conveyor, or lower the position of the infeed control phototube, or decrease the dispersion weight. 3.For products with large piece weight, increase the upper weight limit.

Table 11-2

Error Name/ Error Description	Cause	Action
[OVERSCALE ERROR] /A weight exceeding TW + UL is detected.	1.Excessive product fed into weigh hopper. 2.Infeed position from the infeeder is one-sided. 3.Position of the infeed control phototube is not proper. TIP - This error occurs when weighted product over "target weight + upper weight limit" or "set target count + upper count limit" is supplied to one head, and when the number of overscale heads becomes more than the set value. If an overscale head exists during operation, an overscale error will occur when the operation is stopped, even if the number of overscale heads is less than the set value. Also for full scale heads (when the weighted product exceeding the weight limit is supplied to one head), an overscale error will occur when the operation is stopped. - If overscale is set to be automatically discharged, production will not be stopped due to this error, and a warning message will be displayed on the lower side of the screen of the remote control unit. However, for full scale heads that do not discharge products automatically during operation, an overscale error will occur when the operation is stopped.	1.Decrease the feeder amplitude or time by making feeder adjustment. 2.Adjust the position of the infeeder (supply chute) so that the infeed position comes to the center of the dispersion table. 3.Too many products are pooled in the dispersion unit because the position of the infeed control phototube is too high. Lower the position of the infeed control phototube. Refer to "Table 11-13 ".

Table 11-2

Error Name/ Error Description	Cause	Action
ERROR /STEPPING MOTOR DRIVE ERROR Hopper does not open/ close properly (:PH, WH, RS, TH, and DTH).	1.Product is jammed. TIP This error occurs in the event of the hopper malfunction in cases that the products are jammed in the hopper or the motor does not operate properly due to excessive load.	1.Remove the jammed products from the hopper. 2.Inspect for looseness, breakage or loss of the screws of the hopper. If unable to repair, contact the distributor or Ishida customer support to request repair service. 1.If the error occurs again after the error is once cleared, check the timing. If this error still occurs, turn off the main power switch and turn it on again. Refer to "Table 11-13 ".
ERROR /POWER ERROR There is no power supplied to the motor (:PH, WH, RS, TH, and DTH).	1.The fuse in the main body is blown. TIP This error occurs when the power is not supplied to the drive unit.	1.Replace the fuse.
ERROR /DRIVE PATTERN CALC ERROR (:PH, WH, RS, TH, and DTH)	1.Setting of the hopper action parameter is improper.	1.Initialize the setting of the drive specification.
ERROR /DRIVE PARAMETER NON-RECEPTION ERROR (:PH, WH, RS, TH, and DTH)	1.Communication is interfered due to momentary power loss of the power supply (A failure in the communication line inside the weigher). 2.The drive unit was requested to drive before receiving the drive parameters from the remote control unit.	1.Turn off the main power switch and turn it on again.
ERROR /CYCLE OVERLAP ERROR (:PH, WH, RS, TH, and DTH)	1.Hopper opening and closing are overlapped. TIP This error occurs when attempting to operate the running drive unit of TH, DTH or RS in the next cycle.	1.The setting of the weigher speed is too high. Lower the weigher speed.

Table 11-2

Error Name/ Error Description	Cause	Action
[PIECE WEIGHT ERROR] /Please check current piece weight.	1.The updated piece weight exceeds the piece weight revision range. TIP This error occurs when the count set weighing (optional) is used.	1.The setting of the weigher speed is too high. Lower the weigher speed.

11.2.2 Pattern 2

If an error occurs during operation, draining, or zero adjustment, warning alarm sounds and the weighing machine stops automatically.

In addition, error name, error location, error description and result, cause of the error and its action are displayed on the screen of the remote control unit.

When a pattern 2 error occurs, refer to "Table 11-3 " to eliminate the cause of the error and clear the error.

Table 11-3

Error Name/ Error Description	Cause	Action
[ZERO ERROR]	1.Weighed product is remaining in weigh hopper. 2.Weighed product is spilling from the pool hopper. TIP This error occurs when the reading is not within the permissible range. This error also occurs when extremely heavy hoppers are mounted or excessive load is applied to the cell in the plus or minus direction.	After checking the following points, perform the manual zero adjustment. 1.Remove the products remaining in the weigh hopper. 2.Adjust the open/close timing of the hopper. Inspect for looseness, breakage or loss of the screws of the weigh hopper. If unable to repair, contact the distributor or Ishida customer support to request repair service. 3.If repair cannot be achieved by the above actions 1 and 2, there is a possibility of malfunctioning of the ADC board or power. Contact the distributor or Ishida customer support.

Table 11-3

Error Name/ Error Description	Cause	Action
[SPAN ERROR]	1.Span adjustment weight is not ***g. TIP This error occurs when the reading is not within the permissible range due to an improper span adjustment weight.	After checking the following points, perform the span adjustment. 1.Place a proper span adjustment weight. Standard capacity specification:200 g 2.If repair cannot be achieved by the above action 1, there is a possibility of malfunctioning of the ADC board or power. Contact the distributor or Ishida customer support to request repair service. Refer to "Table 11-13".
[NETWORK ERROR] /DUC CIRCUIT DISCONNECTION ERROR	1.Error has occurred in the communication network. TIP This error occurs when the weigher is affected by external noise.	1.There is a possibility that the CPU or board is damaged. Contact the distributor or Ishida customer support to request repair service.
[NETWORK ERROR] /DUC NODE SETTING ERROR	1.Error has occurred in the communication network. TIP This error occurs when the CPU or board is damaged.	1.Contact the distributor or Ishida customer support to request repair service.
[RECHECK ERROR]	1.The result of the combination by count is not within "target weight-lower weight limit" and "target weight + upper weight limit". TIP This error occurs when the count set weighing (optional) is used.	1.Check the set value of the upper and lower weight limits.

Table 11-3

Error Name/ Error Description	Cause	Action
[FDRV ERROR]	1.Error has occurred in the communication network. TIP This error occurs when there is a failure in the communication line.	1.Contact the distributor or Ishida customer support.

11.2.3 Pattern 3

If an error occurs during operation, draining, or zero adjustment, warning alarm sounds and the error name is displayed.

When a pattern 3 error occurs, refer to "Table 11-4 " to perform the correcting action. When the action is completed, the error name disappears.

Table 11-4

Error Name/ Error Description	Cause	Action
[WARNING:NO TRANSMISSION]	1.Poor connection in the communication line between the remote control unit and the main unit (Unable to transmit data from the remote control unit to the main unit).	1.Check that the communication line between the remote control unit and the main unit is properly connected. 2.There is a possibility that the ADC, FDC or WCU board is damaged. Contact the distributor or Ishida customer support. Refer to "Table 11-13"
[WARNING:NO RESPONSE]	1.Poor connection in the communication line between the remote control unit and the main unit (The remote control unit was unable to receive the communication data transmitted from the main unit every second for more than 5 seconds). 2.The power is not being supplied to the WCU.	1.Check that the communication line between the remote control unit and the main unit is properly connected. 2.There is a possibility that the WCU board is damaged. Contact the distributor or Ishida customer support.
[LOW PRODUCT]	1.The product counts on the dispersion table are small.	1.Supply products to the dispersion table. 2.If continued operation with low products is desired, set the [Infeed Control] key to OFF.

Table 11-4

Error Name/ Error Description	Cause	Action
[WARNING:OVERSCALE]	1.This error occurs when weighted product over "target weight + upper weight limit" or "set target count + upper count limit" is supplied to one head. A warning message will be displayed only when the overscale head count is less than the set head count or the head count is set to "0".	1.Decrease the feeder amplitude or time by making feeder adjustment. 2.Decrease the amount of products supplied to the dispersion table (Decrease the dispersion weight).
[WARNING:DETACHED HEAD]	1.The weigh hopper is not installed. 2.Zero adjustment was performed with a product stuck to the weigh hopper, and the weighted product came out of the weigh hopper.	1.Make sure that the weigh hopper is properly installed. 2.Check that there is no product stuck to the weigh hopper, then perform zero adjustment.
[WARNING:OVERWEIGHT ERROR]	1.A proper result of combination weighing cannot be obtained, and all combination weights are underweight or overweight. 2.During rechecking, overweight occurs due to vibration, etc. TIP This error is displayed when overweight dump count is less than the set count or the dump count is set to "0".	1.Decrease the feeder amplitude or time by making feeder adjustment. 2.Decrease the amount of products supplied to the dispersion table. 3.Increase the set value of the upper weight limit.
[RCU fan stopped. Contact to factory service.]	1.The RCU fan connector is disconnected. 2.The RCU fan is broken.	1.Turn off the main power switch to avoid damaging the boards. Contact the distributor or Ishida customer support to request repair service.
[WARNING:AUTO ZERO ERROR]	1.The weigh hopper is constantly loaded.	1.Check the weigh hopper. 2.There is a possibility of malfunctioning of the weight sensor. Contact the distributor or Ishida customer support.
[WARNING:COMMUNICATION ERROR]	1.Error has occurred in the communication network.	1.Contact the distributor or Ishida customer support to request repair service.

Table 11-4

Error Name/ Error Description	Cause	Action
[WARNING:RECHECK ERROR]	1.The result of the combination by count is not within "target weight-lower weight limit" and "target weight + upper weight limit". TIP This error occurs when the count set weighing (optional) is used.	1.Check the set value of the upper and lower weight limits.
ERROR /STEPPING MOTOR DRIVE ERROR Hopper does not open/ close properly. (;PH and WH)	1.Product is jammed. TIP This error occurs in the event of the hopper malfunction in cases that the products are jammed in the hopper or the motor does not operate properly due to excessive load.	1.Remove the jammed products from the hopper. 2.Inspect for looseness, breakage or loss of the screws of the hopper. If unable to repair, contact the distributor or Ishida customer support to request repair service. 3.If the error occurs again after the error is once cleared, check the timing. If this error still occurs, turn off the main power switch and turn it on again. Refer to "Table 11-13 ".
ERROR /POWER ERROR There is no power supplied to the motor. (;PH and WH)	1.The fuse in the main body is blown. TIP This error occurs when the power is not supplied to the drive unit.	1.Replace the fuse.
ERROR /DRIVE PATTERN CALC ERROR (;PH and WH)	1.Setting of the hopper action parameter is improper.	1.Initialize the setting of the drive specification.

Table 11-4

Error Name/ Error Description	Cause	Action
ERROR /DRIVE PARAMETER NON-RECEPTION ERROR (.:PH and WH)	1.Communication is interfered due to momentary power loss of the power supply (A failure in the communication line inside the weigher). 2.The drive unit was requested to drive before receiving the drive parameters from the remote control unit.	1.Turn off the main power switch and turn it on again.
ERROR /CYCLE OVERLAP ERROR (.:TH, DTH, and RS)	1.Hopper opening and closing are overlapped. TIP This error occurs when attempting to operate the running drive unit of TH, DTH or RS in the next cycle.	1.The setting of the weigher speed is too high. Lower the weigher speed.

11.2.4 Pattern 4

When an error occurs in data when opening the data from memory or USB, an error message is displayed.

In addition, an error screen is displayed on the screen of the remote control unit.

When a pattern 4 error occurs, press the [Exit] key to exit the error screen and refer to "Table 11-5 " to perform the correcting action.

Table 11-5

Error Name/ Error Description	Cause	Action
[The memory of <Product Set> may be corrupt.]	1.The internal data is damaged.	1.Initialize the memory.
[The memory of <Weigh Spec. Set> may be corrupt.]	1.The internal data is damaged.	1.Initialize the memory.
[The memory of <Combi.Calc.Spec.Set> may be corrupt.]	1.The internal data is damaged.	1.Initialize the memory.
[The memory of <Sectioning Method Setting> may be corrupt.]	1.The internal data is damaged.	1.Initialize the memory.

Table 11-5

Error Name/ Error Description	Cause	Action
[The memory of <Infeeder Control SET> may be corrupt.]	1.The internal data is damaged.	1.Initialize the memory.
[The memory of <Packer Interlock Setting> may be corrupt.]	1.The internal data is damaged.	1.Initialize the memory.
[COPY/Initialize error]	1.The data in the USB is damaged.	1.Initialize the USB.

11.2.5 Pattern 5

If an error occurs during operation or draining, warning alarm sounds and the weighing machine stops automatically and the power is turned off automatically.

In addition, an error screen is displayed on the screen of the remote control unit.

Refer to "Table 11-6 " for details on the error screen displayed in Pattern 5.

When a Pattern 5 error occurs, press the [Exit] key to exit the error screen and refer to "Table 11-6 " to perform the correcting action.

Table 11-6

Error Name/ Error Description	Cause	Action
[AFV ZERO ERROR]	1.When performing manual zero adjustment or auto zero adjustment of AFV cell, zero cannot be obtained.	1.Perform manual zero adjustment.
[AFV PARAMETER ERROR]	1.Improper weighing data.	1.Check the weigher specification setting.
[AFV COMPENSATION ERROR]	1.AFV compensation is not operating properly.	1.Contact the distributor or Ishida customer support to request repair service.
[AFV OUTPUT ERROR]	1.Poor connection of the connector. 2.Contact with the spacers. 3.Improper sensitivity of the AFV cell.	1.Contact the distributor or Ishida customer support to request repair service.
[AFV OVERFLOW ERROR]	1.Poor connection of the connector. 2.Contact with the spacers. 3.Improper sensitivity of the AFV cell.	1.Contact the distributor or Ishida customer support to request repair service.

11.2.6 Pattern 6

If an error occurs during operation, weighing machine stops automatically.

In addition, an error screen is displayed on the screen of the remote control unit.

When a Pattern 6 error occurs, press the [Exit] key to exit the error screen and refer to "Table 11-7 " to perform the correcting action.

Table 11-7

Error Name/ Error Description	Cause	Action
[Total Overflow!]	1. There is not enough memory for total data backup.	1. Clear all of the totals or change the "Action for Statistical Mem. Full" from "Stop" to "Continue" in order to continue the operation.

11.2.7 Pattern 7

If an error occurs during operation or draining, the weighing machine stops automatically and the power is turned off automatically.

In addition, an error screen is displayed on the screen of the remote control unit.

When a Pattern 7 error occurs, refer to "Table 11-8 " to perform the correcting action.

If the error still occurs, contact the distributor or Ishida customer support to request repair service.

Table 11-8

Error Name/ Error Description	Cause	Action
[ADC firmware error]	1. System error.	1. Turn off the main power switch and turn it on again. 2. If repair cannot be achieved by the above action 1, contact the distributor or Ishida customer support to request repair service.

11.2.8 Pattern 8

When the file transfer fails, warning alarm sounds.

In addition, an error screen is displayed on the screen of the remote control unit.

When a Pattern 8 error occurs, refer to "Table 11-9 " to perform the correcting action.

Table 11-9

Error Name/ Error Description	Cause	Action
FILE TRANSFER ERROR	1. The ftp server is down. 2. The network is disconnected.	1. Contact the network administrator.

11.2.9 Pattern 9

The error message is displayed if CF is not inserted into DMU board, or there are no master file in DMU. When a Pattern 6 error occurs, press the [Exit] key to exit the error screen and refer to "Table 11-10 " to perform the correcting action.

Table 11-10

Error Name/ Error Description	Cause	Action
[Can not access Master Data!]	1.CF is not inserted into DMU board.	1.Inert DF into DMU board.
[No data in Master Memory]	1. There are no master file in DMU.	1.Contact the distributor or Ishida customer support to request repair service.

11.2.10 Pattern 10

The error message is displayed if there is any problem in RCU fan trouble occurs. When a Pattern 6 error occurs, press the [Exit] key to exit the error screen and refer to "Table 11-11 " to perform the correcting action.

Table 11-11

Error Name/ Error Description	Cause	Action
[RCU fan stopped. Contact to factory service.]	1.The RCU fan connector is disconnected. 2.The RCU fan is broken.	1.Turn off the main power switch to avoid damaging the boards. Contact the distributor or Ishida customer support to request repair service.

11.2.11 Error Clear and Action

When an error occurs, follow the procedure described below to eliminate the cause of the error and to clear the error display.

This section describes the [Power] key, [Err Clr & Stop] key and [E Clr & Restrt] key displayed in the error screen of patterns 1 and 2.

These keys are used to clear the errors of patterns 1 and 2 as shown below.

Error clear and action

Clearing error by using the [E Clr & Restrt] key:

Method for restarting the operation after removing the cause of the error.

Clearing error by using the [Err Clr & Stop] key:

Method for stopping the operation after removing the cause of the error.

Clearing error by using the [Power] key:

Method for forcibly clearing the error when the cause of the error cannot be removed or when the error cause is unknown and the error repeatedly occurs even after the error clear operation is performed.

11.3 Failure and Malfunction of the Weighing Machine

When the weighing machine fails but no error message is displayed, perform the corrective action by referring to "Table 11-12".

DANGER

- All inspection and servicing of the main body, control panel or operation panel can be performed only by Ishida service department. Contact the distributor or Ishida Customer Support.

Table 11-12

Status	Cause	Action
The main power breaker shuts down.	Electric leakage of the weighing machine.	1.Contact the distributor or Ishida customer support. Refer to "Table 11-14 "
	Short circuit in the electrical circuit of the weighing machine.	
The products are discharged at a timing other than the discharging.	Foreign matter or product is caught in the pool hopper, weigh hopper or timing hopper.	1.Remove the obstruction.
Nothing is displayed on the screen of the remote control unit.	The main switch is not turned on.	1.Turn on the main switch (Customer's equipment).
	Power switch is turned OFF.	1.Turn ON the breaker (PS-0 unit). 2.Turn ON the circuit protector of the PS-0 unit.
	Breaking or poor connection.	1.Contact the distributor or Ishida customer support. Refer to "Table 11-14 "
	Chattering when turning on the power.	1.Turn OFF the power and turn ON again.
	Display malfunction.	1.Contact the distributor or Ishida customer support. Refer to "Table 11-14 "

Table 11-12

Status	Cause	Action
The weigher does not operate when the power of the remote control unit is turned ON.	RCU board malfunction.	1.Contact the distributor or Ishida customer support. Refer to "Table 11-14 "
	The interlock signal is not input. Interlock between the relay units. LED for switch monitor does not blink (abnormal).	1.Turn on the packer. 2.Make sure that the interlock signal cable is not broken (Is the interlock signal DC24 V input to the external connection terminal?). Refer to "Table 11-14 "
	The interlock signal is not input. Interlock between the relay units. LED for switch monitor blinks (normal).	1.Contact the distributor or Ishida customer support. Refer to "Table 11-14 "
	FDC board malfunction. The hourglass continues to rotate and the [Power] key cannot be set to ON.	1.Contact the distributor or Ishida customer support. Refer to "Table 11-14 "
Underweight occurs frequently.	The amount of product supplied to a hopper is insufficient.	1.If 6-9 heads are selected for combination, the amount of supplied product is insufficient. In this case, increase the feeder amplitude or time. 2.Increase the set value of the dispersion weight.
	The amount of product supplied to the dispersion unit is insufficient.	1.Increase the amount of products supplied from the infeed conveyor and raise the position of the infeed control phototube in order to prevent the shortage of the product supply.
	Affected by vibration.	1.Reinforce the scaffold. 2.Increase the compensation value.
	The speed is too high.	1.The set speed is too high for the product type and volume. Decrease the speed.

Table 11-12

Status	Cause	Action
The weight of the product is significantly different from the displayed weight.	The products are spilling from the hopper.	1.Clean the hopper.
	The weigh hoppers make contact with an object.	1.Adjust the open/close timing of the hopper (PH, WH).
	A product is stuck to the chute or the hopper.	1.Eliminate the object. 2.Clean the chute and the hopper.
	Improper interlock timing with the packer.	1.Adjust the timing.
	The products pass through the hopper.	1.Check the hopper timing adjustment set at the remote control unit.
	The setting of the head specification switch on the DUC board is incorrect.	1.Contact the distributor or Ishida customer support. Refer to "Table 11-14 "
The feeder does not operate.	The fuse is blown.	1.Contact the distributor or Ishida customer support. Refer to "Table 11-14 "
	Board malfunction.	1.Contact the distributor or Ishida customer support. Refer to "Table 11-14 "
	KM01 (electromagnetic contactor) is not operating.	1.Contact the distributor or Ishida customer support. Refer to "Table 11-14 "
The line of a part or the entire display lights up or goes out.	LCD unit malfunction.	1.Contact the distributor or Ishida customer support. Refer to "Table 11-14 "
The display is blank.	LCD unit malfunction.	1.Contact the distributor or Ishida customer support. Refer to "Table 11-14 "
	LCD power malfunction.	1.Contact the distributor or Ishida customer support. Refer to "Table 11-14 "
	RCU power malfunction.	1.Contact the distributor or Ishida customer support. Refer to "Table 11-14 "
The content of the data memory is deleted or changed.	Battery charge failure.	1.Charge the battery.
	RCU board malfunction.	1.Contact the distributor or Ishida customer support. Refer to "Table 11-14 "
	DMU board malfunction.	1.Contact the distributor or Ishida customer support. Refer to "Table 11-14 "

Table 11-12

Status	Cause	Action
The displayed weight data is not stable.	Affected by wind.	1.Prevent the weigher from being directly blown by the wind (It is recommended to perform zero adjustment for all heads and leave the display screen as is).
	Affected by vibration.	1.Reinforce the scaffold, etc.
	Load cell malfunction.	1.Contact the distributor or Ishida customer support. Refer to "Table 11-14 "
	Preamp board malfunction.	1.Contact the distributor or Ishida customer support. Refer to "Table 11-14 "
	ADC board malfunction.	1.Contact the distributor or Ishida customer support. Refer to "Table 11-14 "
The displayed weight data is the maximum or minimum value of the range.	ADC board malfunction.	1.Contact the distributor or Ishida customer support. Refer to "Table 11-14 "
Unintelligible letters are printed on the data sheet, or the data sheet cannot be printed.	RCU board malfunction.	1.Contact the distributor or Ishida customer support. Refer to "Table 11-14 "
	DMU board malfunction.	1.Contact the distributor or Ishida customer support. Refer to "Table 11-14 "
	Printer control board malfunction.	1.Contact the distributor or Ishida customer support. Refer to "Table 11-14 "
	Printer malfunction.	1.Contact the distributor or Ishida customer support. Refer to "Table 11-14 "
Reading from or writing to the USB cannot be performed.	RCU board malfunction.	1.Contact the distributor or Ishida customer support. Refer to "Table 11-14 "
	USB malfunction.	1.Replace the USB.
Condensation is seen on the inside of main body (WP specification).	The dry air supply is stopped.	1.Check the air supply power. Supply dry air to the inside of the main body immediately. Do not turn on the main power until the inside of main body is dried out.

11.4 Troubleshooting

This section describes corrective action to take at the time when an error, failure or malfunction occurs, and corrective action is targeted for maintenance engineers or Ishida service engineers who manage the weighing machine.

DANGER

- **All inspection and servicing of the main body, control panel or operation panel can be performed only by Ishida service department. Contact the distributor or Ishida Customer Support.**

<Trouble with error display>

Table 11-13

Status	Cause	Action
[WARNING:NO TRANSMISSION] (error display)	Connector breaking or poor connection.	Check the conductivity of the following connectors: • J350 (ADC board P-5576*) • XT601 (PS-0 unit) • XJ455 (relay board P-5506*) • XJ453 (relay board P-5506*) • XJ384 (DMU board P-5562*)
[WARNING:NO RESPONSE] (error display)	Connector breaking or poor connection.	Check the conductivity of the following connectors: • XJ384 (DMU board P-5562*)
[ZERO ERROR] (error display)	Board malfunction.	• Replace the ADC board (P-5576*). (When zero is unable to be obtained even when J320 to J322 are disconnected) • Replace the preamp board (P-5527*).
	Weigh mechanism malfunction.	• Replace the load cell.
	±15 V power malfunction.	• Check the ADC board (P-5576*) J324 connector. • Replace the PS-CAL unit.
[SPAN ERROR] (error display)	Board malfunction.	• Replace the ADC board (P-5576*).

Table 11-13

Status	Cause	Action
Hopper Error (PH, WH) (error display)	There is a loose screw on the cam sensor inside the weigh drive unit.	• Repair the weigh drive unit.
	Board malfunction.	• Replace the cam sensor board (P-5207*). • Replace the DUC board (P-5428*/P-5579*/P-5524*).
	Connector breaking or poor connection.	• Check the conductivity of the DUC board (P-5428*/P-5579*/P-5524*) connector.
	The fuse in the main body is blown (PS-0 unit).	• Replace the midget fuse on the DC fuse board (P-5508*) (PS-0 unit) (125 V 5 A).
	Weigh drive unit malfunction.	• Repair or replace the weigh drive unit.
Hopper Error (RS,TH,DTH) (error display)	There is a loose screw on the cam sensor inside the drive unit.	• Repair the drive unit.
	Board malfunction.	• Replace the cam sensor board (P-5207*). • Replace the DUC board (P-5428*/P-5579*/P-5524*).
	Connector breaking or poor connection.	• Check the conductivity of the DUC board (P-5579*/P-5524*) connector. • Check the WCU board (P-5428*/P-5579*/P-5524*).
	The fuse in the main body is blown. (PS-0 unit)	• Replace the midget fuse on the DC fuse board (P-5508*) (PS-0 unit). (125 V 5 A)
	Drive unit malfunction.	• Repair or replace the drive unit.
[OVERSCALE ERROR]	Infeed control phototube malfunction.	• Replace the phototube to avoid pooling of the product caused by not being able to control the amount of product supplied to the feeder.

<Trouble without error display>

Table 11-14

Status	Cause	Action
The main power breaker shuts down.	Electric leakage of the weighing machine.	• Check for electric leakage and short circuit.
	Short circuit in the electrical circuit of the weighing machine.	
Nothing is displayed on the screen of the remote control unit.	Breaking or poor connection.	Make sure that there is no breaking or poor connection of the following: • XC21 connector (PS-RCU unit). • XJ1 connector (Remote control BOX unit).
	Display malfunction.	• Replace the switching power (PS-RCU unit U01). • Replace the RCU board. • Replace the LCD.
The weigher does not operate when the power of the remote control unit is turned ON.	RCU board malfunction.	• Replace the RCU board.
	Component malfunction.	• Replace the KM01 (electromagnetic contactor). (PS-0 unit)
	The fuse is blown.	• Replace fuses f601 through f604 of the AC fuse board (P-5507*). Standard machine midget fuse (250 V 3.15 A)(PS-0 unit).
	The interlock signal is not input. Interlock between the relay units. LED for switch monitor does not blink (abnormal).	• Replace the relay board (P-5506*).
	The interlock signal is not input. Interlock between the relay units. LED for switch monitor blinks (normal).	• Replace the relay board (P-5506*). • Check the connection between the relay board (P-5506*) XJ454 and WCU board (P-5561*) J305.
	FDC board malfunction. The [Power] key cannot be turned ON.	• Replace the FDC board (P-5532*).
The weight of the product is significantly different from the displayed weight.	The setting of the head specification switch on the DUC board is incorrect.	• Check the test drive and set correctly by SW1 on the DUC board (P-5428*/P-5579*/P-5524*). (DUC unit)

Table 11-14

Status	Cause	Action
The feeder does not operate.	The fuse is blown.	• Replace fuses f601 through f604 of the AC fuse board (P-5507*). Standard machine midget fuse (250 V 3.15 A)(PS-0 unit).
	Board malfunction.	• Replace the FDRV board (P-5578*) or FDC board (P-5532*). (Main electrical unit)
	KM01 (electromagnetic contactor) is not operating.	• Check the relay board (P-5506*), WCU board (P-5561*) and harness.
The line of a part or the entire display lights up or goes out.	LCD unit malfunction.	• Replace the LCD unit.
The display is blank.	LCD unit malfunction.	• Replace the LCD unit.
	LCD power malfunction.	• Replace the LCD power.
	RCU power malfunction.	• Replace the power of PS-RCU unit U01.
The content of the data memory is deleted or changed.	Battery voltage decrease.	• Replace the battery. RCU board (Remote control BOX unit) • Replace the battery. DMU board (P5562*) (CAL unit)
	RCU board malfunction.	• Replace the RCU board. (Remote control BOX unit)
	DMU board malfunction.	• Replace the DMU board (P-5562*). (CAL unit)
The displayed weight data is not stable.	Load cell malfunction.	• Replace the weigh drive unit.
	Preamplifier board malfunction.	• Replace the preamp board (P-5527*). (Weigh drive unit)
	DUC board malfunction	• Replace the DUC board.
	ADC board malfunction.	• Replace the ADC board (P-5576*). (CAL unit)
The displayed weight data is the maximum or minimum value of the range.	ADC board malfunction.	• Replace the ADC board (P-5576*). (CAL unit)

Table 11-14

Status	Cause	Action
Unintelligible letters are printed on the data sheet, or the data sheet cannot be printed.	RCU board malfunction.	• Replace the RCU board (Remote control BOX unit).
	RCU I/F board malfunction	• Replace the RCU I/F board (P-5654*). (Remote control BOX unit)
	DMU board malfunction.	• Replace the DMU board (P-5562*). (CAL unit)
	Printer malfunction.	• Replace the printer (Remote control BOX unit).
Reading from or writing to the USB cannot be performed.	RCU board malfunction.	• Replace the RCU board (Remote control BOX unit).
	RCU I/F board malfunction	• Replace the RCU I/F board (P-5654*). (Remote control BOX unit)
	USB malfunction.	• Replace the USB.
Condensation is seen on the inside of main body (WP specification).	The dry air supply is stopped.	• Check the air supply power. Supply dry air to the inside of the main body immediately. Do not turn on the main power until the inside of main body is dried out.
Color of grains in the dew point checker are turning pink, white or brown (WP specification).	Water and/or oil flows into the membrane air dryer.	• Check the operating condition of separators, and if the drain piping rises or bends. • Check if the element has been replaced properly. • When the color of the dew point checker is brown, replace the dew point checker and module set.
Grains in the dew point checker are crashed (WP specification).		• Replace the dew point checker.

Table 11-14

Status	Cause	Action
Color of grains in the dew point checker is turning pink or white (WP specification).	Inlet air temperature is high.	• Lower inlet air temperature (use conditions:-5 to 50°C; no freezing).
	Ambient temperature is high.	• Improve ventilation to lower the ambient temperature (use conditions:-5 to 50°C; no freezing).
	Air flow rate is large.	• Reduce the flow rate to rated flow rate at reference performance:375 L/min(ALR) or under.
	Inlet air pressure is low.	• Set the air pressure between 0.3 Mpa and 1.0 Mpa.
	Purging air volume is small.	• Check if the outlet for purging air is clogged.